unlike any you have seen before, swirls around and around the hole in a tornado-type motion. This hole must be spinning fast! The axis of spin is easy to identify; it is the axis about which the gas stream swirls. The two jets, you notice, shoot out along the spin axis. They are born just above the horizon's north and south poles, where they suck up energy from the hole's spin and from the doughnut,[34] much like a tornado sucks up dust from the earth.

The contrast between Gargantua and 3C273 is amazing: Why does Gargantua, with its 1000 times greater mass and size, not possess an encircling doughnut of gas and gigantic quasar jets? Bret, after a long telescopic study, tells you the answer: Once every few months a star in orbit around 3C273's smaller hole strays close to the horizon and gets ripped apart by the hole's tidal force. The star's guts, roughly 1 solar mass worth of gas, get spewed out and strewn around the hole. Gradually internal friction drives the strewn-out gas down into the doughnut. This fresh gas compensates for the gas that the doughnut is continually feeding into the hole and the jets. The doughnut and jets thereby are kept richly full of gas, and continue to shine brightly.

Stars also stray close to Gargantua, Bret explains. But because Gargantua is far larger than 3C273, the tidal force outside its horizon is too weak to tear any star apart. Gargantua swallows stars whole without spewing their guts into a surrounding doughnut. And with no doughnut, Gargantua has no way of producing jets or other quasar violence.

As your starship continues to rise out of Gargantua's gravitational grip, you make plans for the journey home. By the time your ship reaches the Milky Way, the Earth will be 4 billion years older than when you left. The changes in human society will be so enormous that you don't want to return there. Instead, you and your crew decide to colonize the space around a spinning black hole. You know that just as the spin energy of the hole in 3C273 helps power the quasar's jets, so the spin energy of a smaller hole can be used as a power source for human civilization.